When Tradable Inflation Becomes Domestic: Second-Round Effects and Monetary Policy Trade-Offs*

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Abstract

Appendix

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1 Introduction

The post-pandemic inflation episode in the euro area was remarkable not only for its initial severity but for its persistence. Leading explanations in the literature point to demand-side pressures from fiscal stimulus and the pandemic-era reallocation of household expenditure, supply-chain disruptions that constrained the production of manufactured goods, and large shocks to energy and food prices (Bernanke & Blanchard, 2025; Dao, Gourinchas, Leigh, & Mishra, 2024). These accounts are well-grounded, yet they share a common limitation: they treat the initial shock as the primary object of interest and leave underexplored the domestic propagation mechanism that determined how long that shock remained embedded in prices. This paper argues that a complementary mechanism was central to the persistence of the episode: strong second-round pass-through from tradable to non-tradable inflation.

Second-round pass-through is the process by which an initial increase in the prices of traded goods propagates into the prices of services and other non-tradable goods. When this channel is weak, an imported cost shock is largely a relative price shift: it raises headline inflation temporarily but leaves the domestic wage and price-setting process undisturbed. When it is strong, the same shock activates domestic inflation dynamics, generates persistence, and alters the trade-offs facing monetary policy in a fundamental way. The policy question is then not only what caused the original shock but whether and how quickly it became embedded in the domestic prices that are directly relevant for the ECB's inflation target.

This paper studies the dynamic causal pass-through from tradable to non-tradable inflation in a monthly panel of 19 euro area countries from January 1998 to December 2025. The European Monetary Union is the natural laboratory for this question. The common monetary policy and shared exchange rate remove the confounding variation in policy responses that complicates cross-country comparisons: differences in pass-through across member states reflect genuine variation in structural conditions rather than differential interest rate or exchange rate reactions. At the same time, the EMU's institutional design creates a specific and consequential policy problem. The ECB sets a single interest rate for economies that differ substantially in how imported cost shocks propagate into domestic prices. When

second-round pass-through is strong in some member states and weak in others, the common stance will be systematically mis-calibrated, too accommodative where amplification is strong and unnecessarily tight where the domestic price-setting process is more insulated. Measuring and understanding this heterogeneity is therefore a precondition for evaluating the welfare costs of asymmetric inflation dynamics inside the union.

Our central data contribution is a new monthly panel of disaggregated HICP components with time-varying ECOICOP expenditure weights, which allows us to construct tradable and non-tradable inflation directly from consumer price data rather than relying on producer prices or import prices as proxies, as much of the existing literature does Ha, Kose, and Ohnsorge (2023). The tradable expenditure share of each country's CPI basket varies across both countries and time, and this variation is the foundation of our identification strategy.

Identification proceeds through a panel local projection instrumental variables (LP-IV) estimator in which tradable inflation is instrumented by a Bartik shift-share $Z_{it} = w_{it} \cdot \varepsilon_t$, where w_{it} is the country-time-varying tradables expenditure share and ε_t is the market-based oil price expectation shock of Baumeister and Hamilton (2019). The shift-share structure means that the same common oil price impulse generates heterogeneous variation in tradable inflation across countries through differences in basket composition, providing identifying variation that is orthogonal to country-specific demand conditions. The baseline results are robust across three instrument variants that exploit different sources of cross-country variation.

The baseline LP-IV impulse response of non-tradable inflation to a one-percentage-point increase in instrumented tradable inflation is positive and statistically significant throughout the thirteen-month estimation window. The response rises from 0.18 at impact to a first peak of 0.93 at month six, moderates to 0.75 at month eight, then recovers to a second peak of 0.93 at month ten before declining to 0.54 at the end of the horizon. This double-humped profile is consistent with two transmission channels operating at different lags: a faster input-cost channel, in which service-sector firms pass through higher material and energy costs within several months, and a slower wage-bargaining channel that re-amplifies the response around month ten.

The more consequential finding is that this pass-through is not a structural constant. Under supply-chain stress, measured by the Global Supply Chain Pressure Index of Benigno et al. (2022), peak pass-through nearly triples relative to normal times, rising from below 0.5 to approximately 1.5. The exchange rate dimension produces the sharpest contrast: depreciation episodes generate a peak pass-through of approximately 2.05 at month ten, roughly four times the modest and statistically insignificant response observed during appreciation episodes, indicating that currency movements are a first-order amplifier of second-round effects rather than a channel through which they dissipate. Elevated inflation volatility similarly amplifies pass-through, consistent with high price instability lowering the threshold at which service-sector firms transmit cost shocks into prices. These three conditions, supply-chain disruptions, exchange rate depreciation, and elevated inflation volatility, mutually reinforced one another during the 2021–22 euro area episode, offering a coherent structural account of why domestic service price inflation remained elevated long after energy prices had peaked.

We bring the empirical estimates to bear on monetary policy by interpreting the estimated pass-through as a structural parameter within a two-sector small open economy New Keynesian model. The model allows us to evaluate how the welfare costs of the ECB's one-size-fits-all stance depend on the underlying degree of second-round pass-through across member states. We find that the costs of policy misalignment are convex in the degree of pass-through: applying a rule calibrated to average conditions in a high-pass-through environment is substantially more costly than the symmetric error in a low-pass-through environment, precisely because strong second-round effects generate persistence that a uniform policy stance cannot offset.

The paper most closely related to ours is Merendino and Monacelli (2025), who document that the pass-through of energy prices to consumer price inflation is state-dependent and amplified during periods of supply chain uncertainty, using state-dependent local projections and a New Keynesian model with firm-level Bayesian learning. The key distinction is the object we study: Merendino and Monacelli (2025) focus on the transmission from energy prices to headline consumer inflation, a channel that operates primarily through the

tradable sector. We focus on the subsequent step, the propagation from tradable to non-tradable prices, which is the margin that determines whether an imported shock becomes embedded in domestic inflation. The two papers are therefore complementary: theirs identifies the conditions under which energy shocks are amplified into aggregate consumer prices, ours identifies the conditions under which those amplified tradable price movements then feed through into the service price dynamics that are most directly relevant for the ECB's target. More broadly, this paper contributes to the literatures on open-economy inflation pass-through (Burstein & Gopinath, 2014; Corsetti, Dedola, & Leduc, 2008), second-round inflation dynamics in the euro area (Bobeica & Jarociński, 2019), the macroeconomic consequences of supply-chain disruptions (Ascari, Bonam, & Smadu, 2024; Benigno et al., 2022), and optimal monetary policy under imported inflation in currency unions (Galí & Monacelli, 2005; Faia & Monacelli, 2008).

The remainder of the paper is organized as follows. Section 3 describes the data construction. Section 4 presents the empirical strategy. Section 5 reports the baseline and heterogeneity results. Section 6 develops the structural model and policy evaluation. Section **??** concludes.

2 Literature Review

The propagation of external cost shocks into domestic non-tradable prices so-called second-round effects is central to monetary policy in open economies, yet the empirical literature on this transmission channel lacks causal identification. Reis and Watson (2010) decompose U.S. consumer prices into relative-price changes and pure inflation, finding that aggregate relative-price movements driven largely by tradable goods account for most inflation variability, but they do not estimate the causal link from tradable to non-tradable components. Altissimo, Mojon, and Zaffaroni (2009) show that heterogeneous propagation of common shocks to euro area services prices is the principal source of aggregate inflation persistence, establishing the non-tradable sector as the locus where transitory cost shocks become persistent price-level shifts. On specific commodity channels, Peersman (2022) demonstrates that international food price shocks trigger second-round effects in the euro area through wages and inflation expectations, with significant cross-country heterogeneity in the wage

response. Bobeica and Jarociński (2019) find that domestic propagation mechanisms not global factors alone explain the euro area's "missing inflation" episode of 2012–2014, pointing to internal transmission as a first-order concern. Most recently, Andriantomanga (2025) documents systematic spillovers from tradable to non-tradable inflation across 40 countries using a TVP-VAR, while the reverse channel remains muted. These studies collectively establish that tradable price shocks feed into services inflation and that this channel drives persistence. However, none provides causal identification of the pass-through coefficient, nor do they construct tradable and non-tradable price indices directly from disaggregated consumer price components rather than relying on coarse goods-versus-services proxies. This paper addresses both limitations.

The debate over the 2021–2023 inflation surge provides further motivation. Bernanke and Blanchard (2025) attribute the U.S. inflation episode primarily to commodity prices and pandemic supply disruptions rather than wage-price spirals, a finding they extend to eleven advanced economies (Bernanke & Blanchard, 2024). Shapiro (2024) decomposes PCE inflation into supply- and demand-driven components and concludes that supply factors accounted for more than half of the 2021–2022 rise. For the euro area, Di Giovanni, Kalemli-Özcan, Silva, and Yildirim (2022) show that sectoral supply bottlenecks and demand shocks, propagating through global production networks, jointly explain the inflation dynamics, while Benigno et al. (2022) introduce the Global Supply Chain Pressure Index and document its predictive power for producer and consumer prices. Hansen, Toscani, and Zhou (2023) decompose euro area price increases and find that import costs and profits not wages accounted for most of the 2022 rise, and Alvarez et al. (2024) confirm that sustained wage-price spirals are historically rare across advanced economies. Cavallo, Lippi, and Miyahara (2024) document that large cost shocks accelerate the frequency of price adjustment, generating faster pass-through than time-dependent models predict. What this body of work largely omits is the propagation mechanism from tradable to non-tradable inflation: existing studies identify initial shocks and quantify aggregate effects but do not isolate the second-round channel through which tradable price increases become embedded in services. Our contribution is to estimate this pass-through causally and to show that its state dependence across supply-chain stress, exchange rate, and inflation-volatility regimes explains why euro area inflation

proved more persistent than standard models predicted.

A related literature on exchange rate pass-through (ERPT) connects directly to our finding that depreciation generates roughly four times the tradable-to-non-tradable pass-through of appreciation. Campa and Goldberg (2005) provide benchmark estimates of partial short-run ERPT into import prices for 23 OECD countries, while Gopinath, Itskhoki, and Rigobon (2010) show that currency of invoicing is a first-order determinant: goods priced in dollars exhibit pass-through of roughly 25 percent versus 95 percent for non-dollar-priced goods. The dominant currency paradigm of Gopinath et al. (2020) formalises how dollar exchange rate movements dominate bilateral rates in pass-through regressions. On asymmetry, Bussière (2013) finds that large depreciations pass through to G7 trade prices disproportionately, and Delatte and López-Villavicencio (2012) use asymmetric cointegration to show that depreciations transmit to consumer prices significantly more than appreciations a result that linear models mask entirely. Carrière-Swallow, Firat, Furceri, and Jiménez (2025) extend the analysis to state dependence, documenting amplified pass-through in high-inflation and high-uncertainty regimes. Our paper shows that this asymmetry extends beyond border prices to the second-round channel: depreciation-driven tradable inflation propagates into non-tradable prices far more strongly than appreciation-driven tradable disinflation.

Methodologically, we combine local projections (Jordà, 2005) with instrumental variables in the LP-IV framework formalised by Stock and Watson (2018), which offers robustness to misspecification advantages over structural VARs (Ramey, 2016). Our Bartik shift-share instrument interacts country-specific expenditure weights on tradable HICP components with common euro area component-level price shocks. Identification follows Goldsmith-Pinkham, Sorkin, and Swift (2020) and Borusyak, Hull, and Jaravel (2022): under the former, validity requires exogeneity of the expenditure shares; under the latter, quasi-random assignment of the price shocks suffices. We exploit oil supply shocks identified following Baumeister and Hamilton (2019) as a component of the shift variation. This design yields plausibly exogenous variation in tradable inflation, enabling the first causal panel estimates of the second-round pass-through to non-tradable prices in the euro area.

3 Data

The data consist of a panel of 19 euro area member states observed at monthly frequency from January 1998 to December 2025, assembled from Eurostat and the ECB Data Portal. Data for Croatia are available from January 1999; all remaining members are observed over the full period, yielding 6,384 country-month observations in total. Table C.1 presents summary statistics. Table C.2 documents data sources. Table C.3 lists country coverage with sample start and end dates and observation counts.

The central data contribution of this paper is the construction of country-specific tradable and non-tradable inflation measures directly from disaggregated HICP components following the European Classification of Individual Consumption according to Purpose (ECOICOP). For each country i and ECOICOP division j, price indices are downloaded from Eurostat dataset `prc_hicp_midx` (monthly, index base $2015 = 100$) and annual expenditure shares from `prc_hicp_inw`. The year-over-year inflation rate for component j is $\pi_{ijt} = 100 \times (P_{ijt}/P_{ij,t-12} - 1)$, where P_{ijt} denotes the price level index. Tradable and non-tradable inflation are then expenditure-weighted averages of component inflation rates within each basket:

$$\pi_{it}^{k} = \frac{\sum_{j \in \|} w_{ijt}\, \pi_{ijt}}{\sum_{j \in \|} w_{ijt}}, \quad k \in \{T, N\} \tag{1}$$

where w_{ijt} is the expenditure share of component j in country i at time t, expanded to monthly frequency by assigning each month the annual weight of its calendar year. The tradable basket $\mathcal{T}$ comprises CP01 (food and non-alcoholic beverages), CP02 (alcoholic beverages and tobacco), CP03 (clothing and footwear), CP05 (furnishings and household equipment), and CP07 (transport). The non-tradable basket $\mathcal{N}$ comprises CP04 (housing, water, electricity, and gas), CP06 (health), CP08 (communication), CP09 (recreation and culture), CP10 (education), CP11 (restaurants and hotels), and CP12 (miscellaneous goods and services). Expenditure shares are normalized to sum to one across the twelve ECOICOP divisions. Most existing work proxies tradable inflation using producer price indices or import prices rather than constructing it directly from disaggregated consumer price components with matching time-varying expenditure weights (Ha et al., 2023); the present

approach follows the HICP-based decomposition of Andriantomanga (2025), who applies the same classification scheme to a broader sample of 40 countries covering the period from January 2000 to August 2022.

The analysis draws on four additional groups of variables. The oil price shock series are from Baumeister and Hamilton (2019): the market-based oil price expectation shock captures forward-looking shifts in the expected path of oil prices, while the structural oil supply shock isolates exogenous disruptions to global crude output identified via sign and magnitude restrictions. Global supply chain conditions enter through the Global Supply Chain Pressure Index (GSCPI) of Benigno et al. (2022), which aggregates shipping costs, airfreight rates, and purchasing managers' index backlogs into a single composite barometer of international logistics disruptions. Domestic cyclical conditions are captured by the seasonally adjusted unemployment rate (percent of the active population) from Eurostat dataset `une_rt_m` and by the year-over-year growth rate of the seasonally and calendar-adjusted manufacturing production index from Eurostat dataset `sts_inpr_m`. External monetary conditions enter through the log monthly change in the ECB nominal effective exchange rate against 40 trading partners, retrieved from the ECB Data Portal API. Import exposure is measured as imports of goods and services as a share of GDP from Eurostat national accounts dataset `nama_10_gdp`, expanded to monthly frequency by carrying forward each annual observation for the corresponding calendar year. Summary statistics, data sources, and country coverage are reported in Tables C.1–C.3.

4 Empirical Strategy

The empirical objective is to identify the dynamic causal pass-through from tradable inflation, π_{it}^T, to non-tradable inflation, π_{it}^N, in a monthly panel of 19 euro area countries. Because tradable and non-tradable inflation comove with common demand conditions and aggregate cost pressures, ordinary least squares conflates pass-through with spurious comovement; identification requires an external source of variation in π_{it}^T that is orthogonal to domestic non-tradable conditions. The instrument is a Bartik shift-share of the form $Z_{it} = w_{it} \cdot Z_t$, where w_{it} is the tradables expenditure share of country i's CPI basket in

month t computed as the sum of ECOICOP expenditure weights for food, clothing, furnishings, transport, and alcohol and Z_t is the market-based oil price expectation shock of Baumeister and Hamilton (2019). Heterogeneity in w_{it} across countries and over time ensures that the same common oil shock translates into proportionally larger cost impulses for countries with larger tradables baskets, generating cross-sectional identifying variation that is unrelated to idiosyncratic domestic conditions. The exclusion restriction requires that Z_{it} affects non-tradable prices only through its effect on tradable inflation; any direct transmission through energy costs in service production is absorbed by twelve lags of both inflation series and the unemployment rate, and country fixed effects absorb permanent cross-country differences in non-tradable price setting. Time fixed effects are excluded because they would absorb Z_t, eliminating the instrument.

[NEED TO ADD MECHANISM TO IDENTIFY REGRESSION HERE]

The dynamic pass-through is estimated horizon by horizon following the local projection method of Jordà (2005). The first stage establishes instrument relevance by regressing tradable inflation on Z_{it} and controls:

$$\pi_{it}^{T} = \alpha_i + \lambda\, Z_{it} + \sum_{\ell=1}^{L} \boldsymbol{\gamma}_\ell' \mathbf{x}_{i,t-\ell} + v_{it}, \tag{2}$$

where α_i are country fixed effects, $Z_{it} = w_{it} \cdot Z_t$ is the Bartik instrument, and $\mathbf{x}_{i,t-\ell} = (\pi_{i,t-\ell}^{N},\, \pi_{i,t-\ell}^{T},\, u_{i,t-\ell})'$ collects lags of non-tradable inflation, tradable inflation, and the unemployment rate. The second stage projects non-tradable inflation h months ahead onto instrumented tradable inflation:

$$\pi_{i,t+h}^{N} = \alpha_i + \theta_h\, \widehat{\pi^{T}}_{it} + \sum_{\ell=1}^{L} \boldsymbol{\gamma}_\ell' \mathbf{x}_{i,t-\ell} + \varepsilon_{i,t+h}, \quad h = 0, 1, \ldots, H, \tag{3}$$

where $\widehat{\pi^{T}}_{it}$ denotes tradable inflation instrumented by Z_{it}. The sequence $\{\theta_h\}_{h=0}^{H}$ traces the dynamic second-round pass-through from tradable to non-tradable inflation, with cumulative pass-through at horizon H given by $\sum_{h=0}^{H} \theta_h$. The main specification uses $H = 13$ monthly horizons and $L = 12$ lags; standard errors are clustered at the country level throughout.

To assess whether pass-through varies across economic states, equation (3) is estimated separately for subsamples defined by regime indicators, an approach that avoids the need for an additional excluded instrument to identify the triple interaction $\widehat{\pi^T}_{it} \cdot D_{it}$. Three regime dimensions are considered. First, the sample is split according to whether the Global Supply Chain Pressure Index (GSCPI) of Benigno et al. (2022) is above or below its standardised long-run mean of zero, separating episodes of supply-chain stress from normal logistical conditions. Second, the sample is split by the sign of the log monthly change in the nominal effective exchange rate, $\Delta \ln \mathrm{NEER}_{it}$, testing for asymmetric pass-through between appreciation and depreciation episodes. Third, the inflation environment is characterised by a 36-month rolling standard deviation of headline HICP inflation, $\hat{\sigma}_{it}$, split at each country's own within-country median rolling standard deviation, $\tilde{\sigma}_i$; periods in which $\hat{\sigma}_{it} > \tilde{\sigma}_i$ are classified as high-volatility, capturing episodes of elevated price instability relative to a country's own historical norm. Each subsample retains the same instrument, controls, and fixed effects as the baseline specification. The full sample spans January 1998 to December 2025, with pre- and post-2020 sub-periods reported to assess parameter stability around the pandemic energy-price episode.

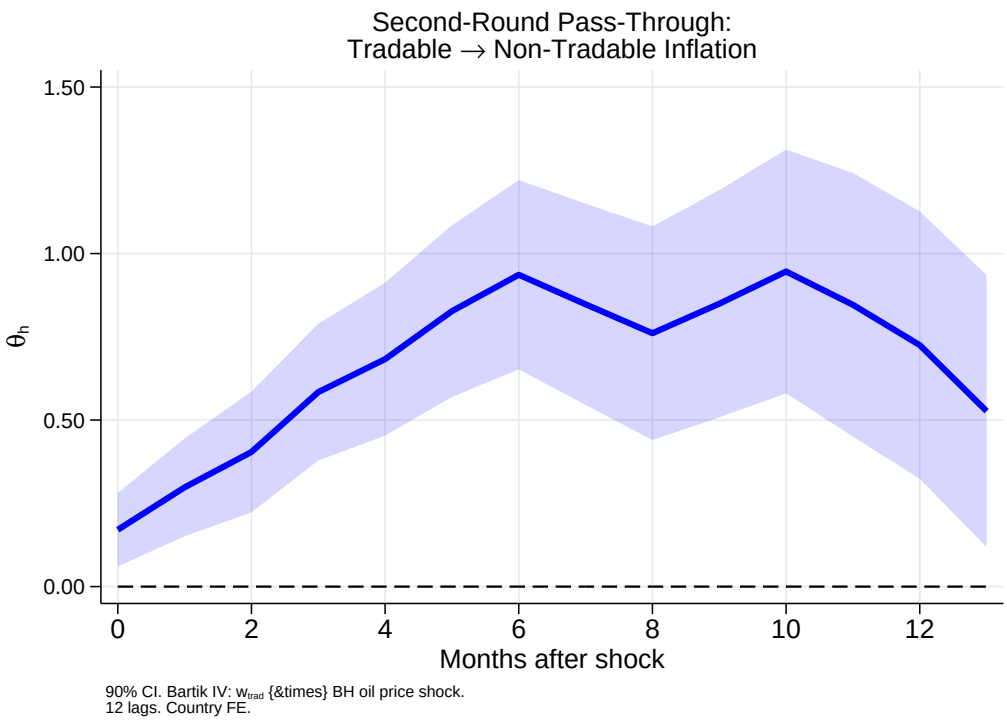
5 Results

5.1 Baseline Pass-Through

Figure 1 reports the LP-IV impulse response of non-tradable inflation to a one-percentage-point instrumented increase in tradable inflation, estimated over the full sample using the Bartik instrument $Z_{it} = w_{it} \cdot Z_t$ with twelve lags and country fixed effects. The response is identified at 90 percent confidence intervals throughout. The point estimates are strictly positive at every horizon, with the 90 percent confidence band lying entirely above zero throughout the thirteen-month window, indicating that second-round pass-through is statistically significant at all horizons. The response rises from approximately 0.18 at impact to a first peak of 0.93 at month six, dips modestly to 0.75 at month eight, recovers to a second peak of 0.93 at month ten, and then declines to 0.54 at the end of the horizon. This double-humped shape suggests that the transmission of tradable cost pressures into

non-tradable prices operates through at least two channels that operate at different lags: a faster channel plausibly direct input-cost pass-through that peaks around month six, and a slower channel consistent with wage bargaining and second-round service-price adjustment that re-amplifies the response around month ten. Cumulatively, the pass-through at twelve months amounts to a substantial and persistent propagation of tradable shocks into non-tradable inflation, consistent with the TVP-VAR evidence in Andriantomanga (2025), who documents positive spillovers from tradable to non-tradable inflation for a broader panel of 40 countries, and providing a causal LP-IV counterpart to that reduced-form result.

Figure 1: Baseline LP-IV Impulse Response of Non-Tradable Inflation



Note: The instrument is the Bartik shift-share $Z_{it} = w_{it} \cdot Z_t$, where w_{it} is the country-time-varying tradables expenditure share and Z_t is the market-based oil price expectation shock of Baumeister and Hamilton (2019). The response is to a one-percentage-point increase in tradable inflation. Specification: 12 lags, country fixed effects, full sample. Shaded band: 90% confidence interval (standard errors clustered at the country level).

5.2 State-Dependent Pass-Through

Whether the second-round pass-through documented above is stable across economic states or amplified in particular conditions is the central question of the regime analysis. Three binary regime indicators partition the estimation sample, and equation (3) is re-estimated on each subsample with the same instrument, controls, and fixed effects as the baseline. Estimating separate LP-IVs within each subsample avoids the need for an additional excluded instrument to identify the interaction term $\widehat{\pi^T}_{it} \cdot D_{it}$.

[NEED TO ADD REGRESSION SPEC HERE]

The first dimension captures global supply-chain conditions through the GSCPI of Benigno et al. (2022):

$$D_t^{\text{GSCPI}} = \mathbf{1}\Big[\text{GSCPI}_t > \tilde{q}\Big],$$

where $\tilde{q}$ denotes the panel median of the GSCPI series; because the index is standardised with mean zero and unit variance, the median threshold is approximately zero, so the high-stress subsample captures episodes of above-normal logistical disruption. The second dimension isolates the direction of the exchange rate:

$$D_{it}^{\text{NEER}} = \mathbf{1}\Big[\Delta \ln \text{NEER}_{it} < 0\Big],$$

where $D_{it}^{\text{NEER}} = 1$ denotes depreciation months, in which imported cost pressures are amplified relative to appreciation episodes. The third dimension measures the inflation volatility environment:

$$D_{it}^{\sigma} = \mathbf{1}\Big[\hat{\sigma}_{it} > \tilde{\sigma}_i\Big],$$

where $\hat{\sigma}_{it}$ is the 36-month rolling standard deviation of headline HICP inflation for country i at time t, and $\tilde{\sigma}_i$ is the within-country median of that rolling standard deviation; high-volatility periods correspond to episodes of elevated price instability relative to a country's own historical norm.

Figure 2 presents the six impulse responses arranged in a 3×2 panel, with the low-regime subsamples on the left and the high-regime subsamples on the right, with 90 percent confi-

dence intervals shown separately for each.

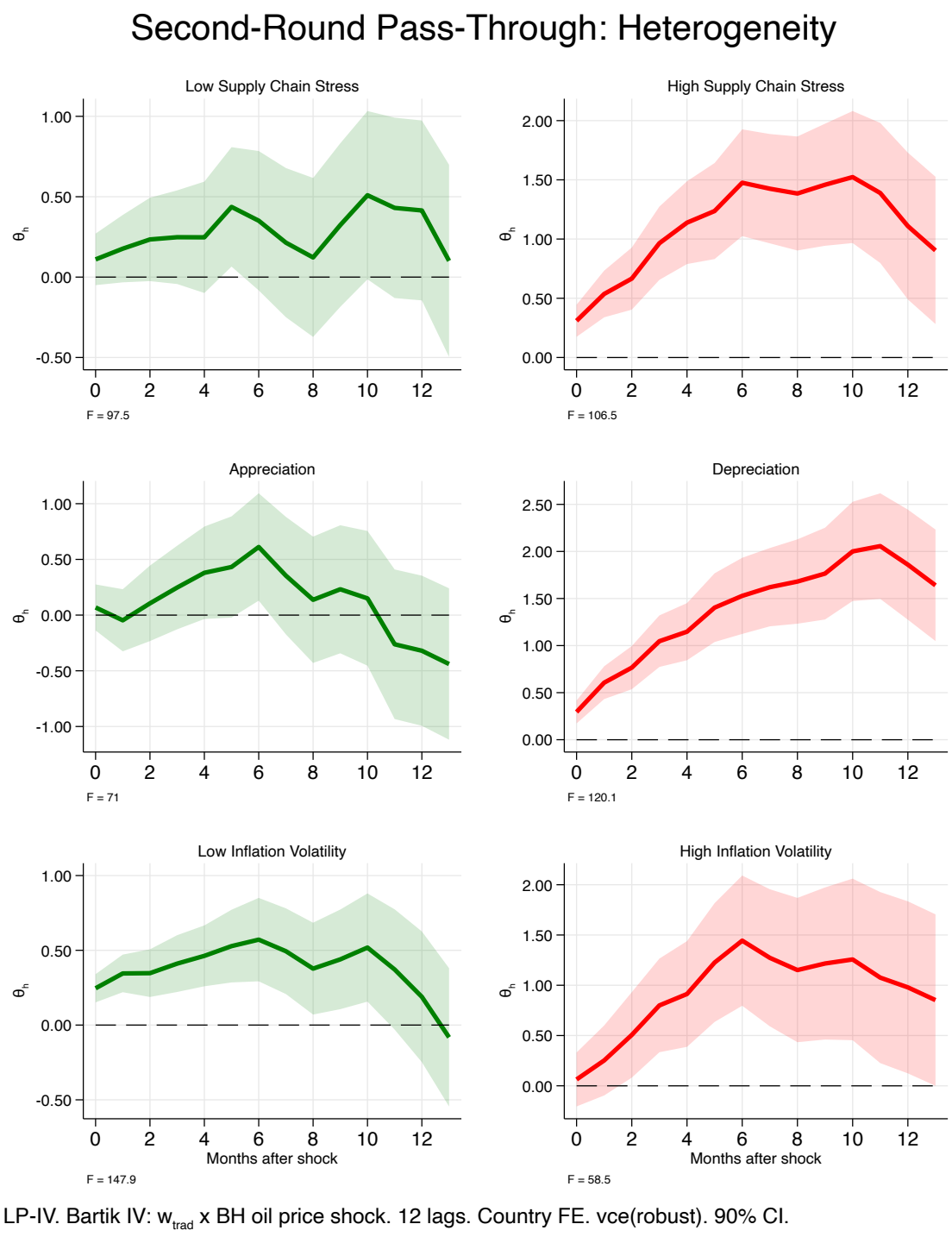
Along the supply-chain dimension, the contrast between the two subsamples is sharp. During low-GSCPI periods, the response oscillates between 0.1 and 0.5 and the confidence band crosses zero at multiple horizons, leaving the pass-through statistically indistinguishable from zero for much of the window. During high-GSCPI periods, the response rises from 0.35 at impact to a plateau of approximately 1.5 between months six and ten, with the 90 percent band lying entirely above zero from month two onward. The near-tripling of peak pass-through under supply-chain stress is consistent with the mechanism that logistics disruptions compress firms' substitution margin between imported and domestic inputs, forcing service-sector producers to absorb a larger share of upstream cost shocks rather than switching suppliers; under normal logistical conditions that margin is wider and cost pressures are more readily diversified away. For monetary policy, this finding implies that a central bank that calibrates its response to second-round effects using normal-times estimates will systematically underreact when global supply chains are under stress.

The exchange rate dimension reveals the most pronounced asymmetry in the dataset. During appreciation episodes ($\Delta \ln \mathrm{NEER}_{it} > 0$), the response is near zero at impact, rises to a modest peak of about 0.55 at month six, and then reverses, falling to approximately -0.5 by months twelve and thirteen; the confidence band crosses zero repeatedly and the overall response is statistically indistinguishable from zero across most of the horizon. During depreciation episodes, the picture is entirely different: the response rises strongly from 0.35 at impact to a sustained peak of approximately 2.05 at month ten, with the confidence band remaining well above zero throughout. The asymmetry is large in magnitude depreciation pass-through at its peak is roughly four times the appreciation peak and carries a clear economic interpretation: when the effective exchange rate depreciates, import costs rise in domestic currency terms, amplifying the cost impulse received by non-tradable producers; when it appreciates, the offsetting import-cost relief attenuates or reverses the propagation. From a monetary policy standpoint, this asymmetry implies that exchange rate depreciation episodes sharply amplify the risk of tradable shocks becoming entrenched in domestic service prices, while appreciation provides a natural buffer against such propagation.

Along the inflation volatility dimension, high-volatility periods display substantially stronger pass-through than low-volatility periods. When inflation volatility is below the within-country median, the response peaks at approximately 0.55 and the confidence band includes zero at several horizons, indicating moderate and imprecisely estimated pass-through. When volatility is above the within-country median, the response rises steeply to a peak of 1.5 at month six and remains above 1.0 through month thirteen, with the confidence band staying above zero throughout. The pattern is consistent with the view that elevated price instability shortens the effective duration of price contracts and lowers the threshold at which service-sector firms choose to adjust prices, so that a given tradable cost impulse translates into a larger and more immediate adjustment in non-tradable prices.

Taken together, the regime results indicate that second-round pass-through is not a stable structural constant but rises sharply when global supply chains are disrupted, when the exchange rate is depreciating, and when the domestic inflation environment is already volatile three conditions that mutually reinforced one another during the 2021–22 inflation episode in the euro area.

Figure 2: State-Dependent LP-IV Impulse Responses of Non-Tradable Inflation



Note: Each panel reports the response to a one-percentage-point increase in instrumented tradable inflation within the indicated subsample. Rows correspond to three regime dimensions: global supply-chain pressure (GSCPI above vs. below panel median), nominal effective exchange rate direction (appreciation vs. depreciation), and inflation volatility (36-month rolling standard deviation of headline HICP above vs. below within-country median). Instrument: Bartik shift-share $Z_{it} = w_{it} \cdot Z_t$. Specification: 12 lags, country fixed effects, full sample. Shaded bands: 90% confidence intervals (standard errors clustered at the country level).

6 Model

7 Conclusions

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When Tradable Inflation Becomes Domestic: Second-Round Effects and Monetary Policy Trade-Offs

Supplemental Appendix

For Online Publication

This appendix contains:

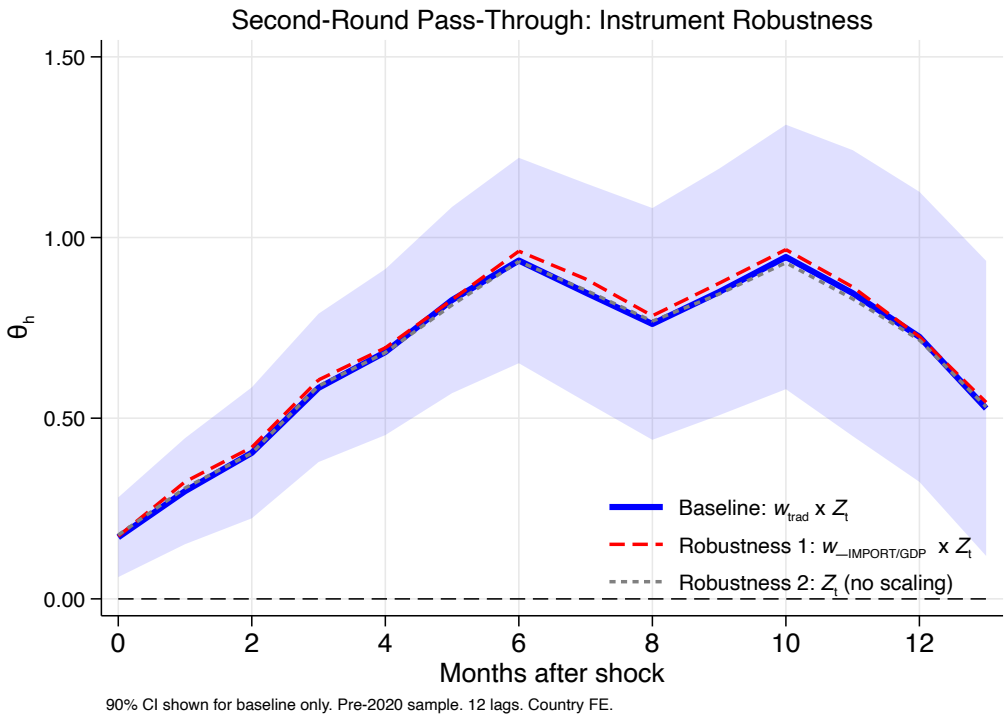
- **Appendix A.** Robustness Checks
- **Appendix B.** Full Model of section 6
- **Appendix C.** Additional Tables and figures

A Appendix A. Robustness

A.1 Robustness

Instrument robustness. Figure A.1 compares the baseline Bartik instrument against two alternatives: a Bartik scaled by import exposure (w_{it}^{imp} = imports/GDP) and the unscaled market-based oil price shock Z_t used directly without country-level weighting. The import-exposure Bartik captures a different dimension of openness, the share of domestic expenditure that is sourced from abroad, while the unscaled instrument exploits only the common time-series variation in Z_t and relies on country fixed effects to absorb permanent differences in exposure. Ninety-percent confidence intervals are shown for the baseline specification only. The three-point estimate paths are virtually indistinguishable across the entire twelve-month horizon: all three rise together from approximately 0.18 at impact to peaks of 0.93 at month six and 0.93 at month ten, with the import-to-GDP Bartik and the unscaled shock tracking the baseline so closely that the lines are nearly superimposed throughout. This near-perfect agreement across instruments that exploit different sources of cross-country heterogeneity, the tradables share of the CPI basket, the aggregate import intensity, and the common time-series variation alone strongly supports the conclusion that the pass-through estimate is not an artifact of any particular exposure weighting and is robust to the choice of how the common oil shock is scaled across countries.

Figure A.1: LP-IV Impulse Responses Under Alternative Instrument Specifications



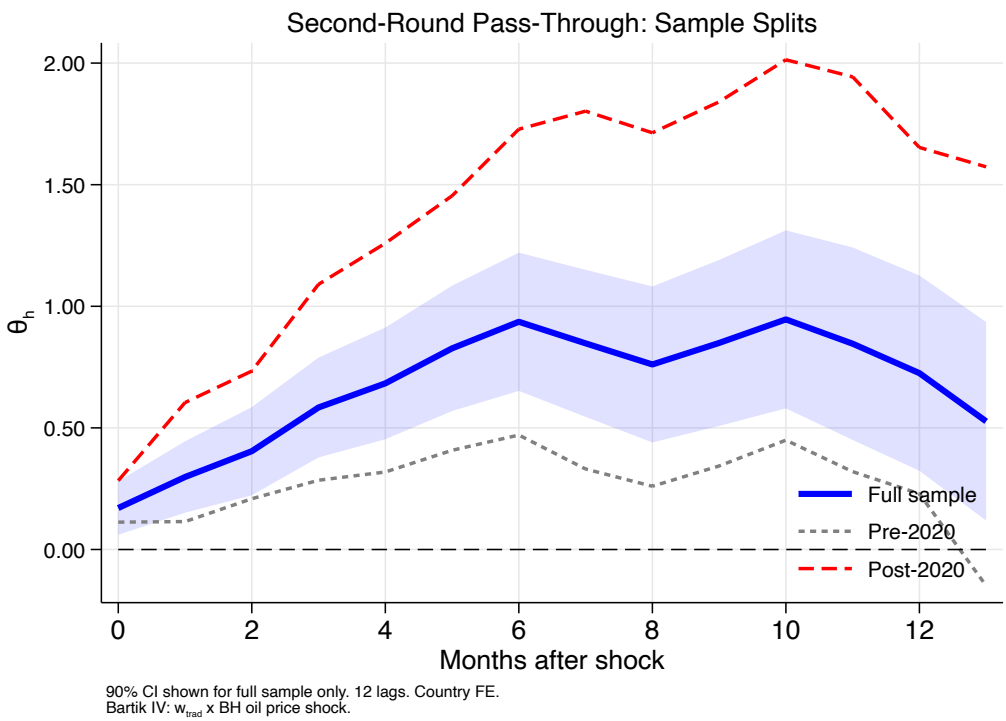
Note: Each line reports the response of non-tradable inflation to a one-percentage-point increase in instrumented tradable inflation under a different instrument specification. Baseline (solid, blue): Bartik instrument $Z_{it} = w_{it} \cdot Z_t$, where w_{it} is the tradables expenditure share. Robustness 1 (dashed, red): Bartik scaled by import-to-GDP ratio, $Z_{it} = (m_{it}/y_{it}) \cdot Z_t$. Robustness 2 (short-dashed, grey): unscaled oil price shock Z_t. Specification: 12 lags, country fixed effects, pre-2020 sample. Shaded band (baseline only): 90% confidence interval.

Sample robustness. Figure A.2 overlays the baseline LP-IV response with pre-2020 and post-2020 sub-period estimates, using the same instrument and controls throughout. The ninety-percent confidence band is shown for the full-sample estimate only. The post-2020 period is characterised by energy price interventions, administered price caps, and supply disruptions that altered the normal relationship between global oil price shocks and euro area tradable inflation.

Figure A.2 reveals a dramatic divergence across sub-periods. The pre-2020 path peaks at approximately 0.47 at month six and fades toward zero by month thirteen, implying a pass-through roughly half the full-sample estimate at its peak. The post-2020 path is strikingly larger: it rises steeply from 0.30 at impact, peaks at approximately 2.02 at month ten, and remains above 1.5 even at the end of the horizon.

The full-sample estimate sits between the two, reflecting the average of a moderate pre-2020 regime and a substantially amplified post-2020 regime. This divergence suggests that the 2020-22 period was characterised by an unusual intensity of second-round propagation, plausibly driven by the confluence of supply-chain disruptions, exchange rate depreciation, and rising inflation volatility documented in Figure 2. The pre-2020 sample therefore, provides the cleaner estimate of the stable, long-run structural pass-through parameter, while the post-2020 results illustrate the conditions under which second-round effects can amplify substantially.

Figure A.2: LP-IV Impulse Responses Across Sample Periods



Note: Each line reports the response of non-tradable inflation to a one-percentage-point increase in instrumented tradable inflation over different sample periods. Full sample (solid, blue), pre-2020 subsample (short-dashed, grey), and post-2020 subsample (dashed, red). Instrument: Bartik shift-share $Z_{it} = w_{it} \cdot Z_t$. Specification: 12 lags, country fixed effects. Shaded band (full sample only): 90% confidence interval.

B Appendix B. Full Model of section 6

C Appendix C. Additional tables and figures

Table C.1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Headline inflation	6,372	2.67	2.85	-4.35	25.16
Tradable inflation	6,372	2.48	3.01	-4.09	22.02
Non-tradable inflation	6,337	2.96	3.30	-7.04	36.63
Market oil price shock	5,776	0.01	0.05	-0.32	0.09
Oil supply shock	6,327	-0.13	1.24	-10.86	3.56
GSCPI	6,384	0.01	1.00	-1.58	4.48
$\Delta \ln$ NEER	6,384	0.10	1.31	-3.92	5.21
Unemployment rate (%)	6,284	8.36	3.97	1.80	26.40
IP growth, YoY (%)	5,568	2.02	8.43	-44.05	79.25
Import exposure (% of GDP)	6,192	61.46	29.61	21.00	189.90

Note: Inflation rates are year-over-year percent changes. GSCPI is standardized with a mean of zero and unit variance. NEER change and oil shocks are in log points. Sample: 19 euro area countries, January 1998 – December 2025.

Table C.2: Data Sources

Variable	Description	Source
`pi_headline`	Headline HICP inflation (YoY, %)	Eurostat `prc_hicp_midx`
`pi_trad`	Tradable inflation, Eq. (1), $\mathcal{B} = \mathcal{T}$	Authors' construction
`pi_nontrad`	Non-tradable inflation, Eq. (1), $\mathcal{B} = \mathcal{N}$	Authors' construction
`w_food`, `w_clothing`, ...	Annual ECOICOP expenditure weights (normalized)	Eurostat `prc_hicp_inw`
`bh_oil_price_exp_shock`	Market-based oil price expectation shock	Baumeister and Hamilton (2019)
`bh_oil_supply_shock`	Structural oil supply shock	Baumeister and Hamilton (2019)
`gscpi`	Global Supply Chain Pressure Index	Benigno et al. (2022)
`dln_neer`	Log monthly change in nominal effective exchange rate	ECB Data Portal API
`ur`	Unemployment rate, seasonally adjusted (% of active population)	Eurostat `une_rt_m`
`ip_gr_yoy`	Manufacturing IP growth, year-over-year (%)	Eurostat `sts_inpr_m`
`wi_imp_gdp`	Imports of goods and services (% of GDP)	Eurostat `nama_10_gdp`

Note: ECOICOP tradable basket $\mathcal{T}$: CP01, CP02, CP03, CP05, CP07. Non-tradable basket $\mathcal{N}$: CP04, CP06, CP08, CP09, CP10, CP11, CP12.

Table C.3: Country Coverage

Country	ISO	Start	End	N
Austria	AUT	1998m01	2025m12	336
Belgium	BEL	1998m01	2025m12	336
Croatia	HRV	1999m01	2025m12	324
Cyprus	CYP	1998m01	2025m12	336
Estonia	EST	1998m01	2025m12	336
Finland	FIN	1998m01	2025m12	336
France	FRA	1998m01	2025m12	336
Germany	DEU	1998m01	2025m12	336
Ireland	IRL	1998m01	2025m12	336
Italy	ITA	1998m01	2025m12	336
Latvia	LVA	1998m01	2025m12	336
Lithuania	LTU	1998m01	2025m12	336
Luxembourg	LUX	1998m01	2025m12	336
Malta	MLT	1998m01	2025m12	336
Netherlands	NLD	1998m01	2025m12	336
Portugal	PRT	1998m01	2025m12	336
Slovakia	SVK	1998m01	2025m12	336
Slovenia	SVN	1998m01	2025m12	336
Spain	ESP	1998m01	2025m12	336

Note: N counts non-missing observations of headline HICP inflation. Sample: 19 euro area member states, January 1998 – December 2025.